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Key Points:

- Physical and hydraulic property magnitudes in mineral soils extended beyond ranges from prior work in permafrost regions in interior Alaska
- Mineral soil thermal properties were substantially affected by gravel content, which is not included in common texture-based estimation
- This provides soil parameter constraints for hypothesis testing and prediction for active layer and permafrost dynamics in upland boreal forests

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Soil Physical, Hydraulic, and Thermal Properties in Interior Alaska, USA: Implications for Hydrologic Response to Thawing Permafrost Conditions

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Abstract Boreal forest regions are a focal point for investigations of coupled water and biogeochemical fluxes in response to wildfire disturbances, climate warming, and permafrost thaw. Soil hydraulic, physical, and thermal property measurements for mineral soils in permafrost regions are limited, despite substantial influences on cryohydrogeologic model results. This work expands mineral soil property quantification in cold regions through soil characterization from the discontinuous permafrost zone of interior Alaska, USA. Values extend beyond the range of prior measurement magnitudes in analogous regions, highlighting the importance of this data set. Rocky and silty upland soil landscape classifications and wildfire disturbance provided guiding frameworks for the sampling and analysis for potential implications for the hydrologic response to thawing permafrost. Bulk density (ρ_b), soil organic matter, soil-particle size distributions (sand, silt, and gravel fractions), and soil hydraulic properties of van Genuchten parameters alpha and N had moderate evidence of differences between silty and rocky classifications. Burned and unburned sites had only moderate evidence of differences for silt fraction. Field-saturated hydraulic conductivity (K_{fs}) was more variable at burned sites compared to unburned sites, which corresponded to observations of greater rooting depths at burned sites and observations of root paths in soil cores for K_{fs} measurement. Soil thermal properties suggested that gravel content may reduce the accuracy of commonly used estimation methods for thermal conductivity. This work provides soil parameter constraints necessary for hypothesis testing and site-specific prediction with cryohydrogeologic models to examine controls on active layer and permafrost dynamics in upland boreal forests.

1. Introduction

There is broad recognition that permafrost conditions in arctic and boreal regions are changing (e.g., Hinzman et al., 2005; Jafarov et al., 2012; Romanovsky et al., 2010; Serreze et al., 2000) in response to gradual “press” disturbances, such as shifting climate (e.g., Hinzman et al., 2005; Osterkamp & Romanovsky, 1999), and more punctuated “pulse” disturbances, such as wildfire (e.g., Brown et al., 2015; Jafarov et al., 2013; Jorgenson et al., 2010; Yi et al., 2009). Permafrost can be defined as ground that is $\leq 0^\circ\text{C}$ for at least two consecutive years. Present seasonal active layer conditions and the future rate and extent of permafrost degradation or aggradation have important hydrologic and biogeochemical consequences. The active layer can be defined as the zone that seasonally freezes and thaws (e.g., Walvoord & Kurylyk, 2016). Impacts of thawing and degrading permafrost include shifts in lateral and vertical water fluxes and changes in water residence time (Jafarov et al., 2018; Lamontagne-Hallé et al., 2018; Lyon & Destouni, 2010; St Jacques & Sauchyn, 2009; Walvoord et al., 2019; Walvoord & Striegl, 2007), which in turn alter fluxes of dissolved organic matter and mineral weathering products (e.g., Koch et al., 2013; Striegl et al., 2005). Permafrost-thaw induced changes in infiltration and drainage influence soil moisture, promoting thermal feedback. These thermal feedbacks result from soil saturation effects on the bulk thermal properties of the shallow subsurface and latent heat fluxes during freezing and thawing, which in turn affects depth to permafrost (Farouki, 1986; Kane et al., 2001; Rawlins et al., 2013; Wright et al., 2009; Yi et al., 2018). Soil saturation also impacts anaerobic to aerobic soil conditions that influence the proportion of soil respiration as CO_2 or CH_4 (Lawrence et al., 2015), which has implications for the strength of the permafrost-carbon feedback. The fate of large stocks of soil organic carbon in high-latitude soils, with an estimated $1,307 \pm 170$ petagrams of soil organic carbon residing in the northern circumpolar permafrost region (Hugelius et al., 2014), is one of the most critical concerns related to permafrost thaw (e.g., Harden et al., 2012; McGuire et al., 2009). Recent

quantification of relatively large stores of mercury in permafrost highlights another concern for thaw-induced mobilization (Schuster et al., 2018). Despite the importance of hydrologic considerations in the transport of carbon and other constituents (e.g., Koch, Runkel, et al., 2013; Striegl et al., 2005; Walvoord et al., 2019) and the known impacts of permafrost thaw on fluxes of water (e.g., Koch et al., 2013; Lyon & Destouni, 2010; Walvoord & Striegl, 2007), predicting linked hydrologic and biogeochemical fluxes in high latitudes remains challenging (Walvoord & Kurylyk, 2016).

Recent cryohydrogeologic modeling efforts have made considerable progress in understanding the most important factors and consequences of changes in permafrost and active layer thickness driven by climate and disturbance shifts. These modeling studies have demonstrated the sensitivity of permafrost table depths to organic layer thickness, vegetation, and snow properties (e.g., Atchley et al., 2016; Briggs et al., 2014; Jafarov et al., 2018; Lamontagne-Hallé et al., 2018; Walvoord et al., 2019); soil thermal properties (e.g., Hinzman et al., 1998); soil saturation (e.g., Chadburn et al., 2015; Rawlins et al., 2013; Subin et al., 2013); soil physical properties such as bulk density and porosity (e.g., Harp et al., 2015; Zipper et al., 2018); soil hydraulic properties such as soil permeability (e.g., Zipper et al., 2018); and parameters such as residual water content (Harp et al., 2015). Recent modeling efforts have simulated unsaturated freeze/thaw dynamics (e.g., Briggs et al., 2014; Lamontagne-Hallé et al., 2018), which requires soil-water retention curve specification. Soil hydraulic properties have added importance in cold regions because one of the most common approaches for representing soil freezing in unsaturated media is the assumption that freezing can be approximated using the drying relation from the soil-water retention curve (Koopmans & Miller, 1966; Kurylyk & Watanabe, 2013; Spaans & Baker, 1996; Walvoord & Kurylyk, 2016). As the scientific community expands toward broader-scale prediction of active layer thickness (Chadburn et al., 2015; Subin et al., 2013) using increasingly sophisticated variably saturated models and surface energy and hydrologic boundary conditions (Lamontagne-Hallé et al., 2018), realistic magnitudes and ranges of mineral soil physical, hydraulic, and thermal properties will be critical for increasing accuracy and reducing uncertainty. The difficulty of obtaining observations needed for establishing and evaluating models in arctic-boreal regions is a primary barrier to accurate representation in global models (Chadburn et al., 2015).

Soil property databases typically used to parameterize regional- and local-scale hydrologic and thermal models are either unavailable or accessible only at very coarse scales in permafrost regions. For example, the STATSGO2 database has limited availability at the 1:1,000,000 scale and is unavailable for many areas in Alaska (Natural Resources Conservation Service, 2018). Until widespread, high-resolution soil property databases are available, one approach for model parameterization in permafrost regions is to divide the landscape into similar soil, hydrologic, and ecologic units using frameworks devised for permafrost areas and then measure soil properties in the framework units to provide property magnitudes and ranges. That approach is followed here for the boreal forest biome of interior Alaska, using two of the four end-members of soil landscapes identified by Jorgenson et al. (2013) based on the ecological land classification framework of Jorgenson et al. (2009, 1999). These soil landscapes were identified based on topography, climate, soil properties, permafrost characteristics, hydrologic regimes, and vegetation composition that have shown characteristic vegetation successional trends following disturbance that impact permafrost conditions (Jorgenson et al., 2013). The two soil landscape end-members focused upon are the rocky and silty upland soil landscapes, which cover approximately 30% and 5% of the areal extent of the boreal forest region of Alaska (Karlstrom, 1964; Nossor et al., 2013) although the 5% estimate for silty uplands is an underestimate because of inaccurate mapping (Jorgenson et al., 2013). Within the field areas, field samples were also taken in regions recently impacted by wildfire disturbance and in regions in an unburned (>100 years) condition.

The primary objectives of this work are as follows:

1. Characterize hydraulic, physical, and thermal properties of near-surface mineral soil within areas of variably discontinuous permafrost in interior Alaska in the rocky and silty upland soil landscapes. This characterization is used to highlight differences between rocky and silty areas in addition to burned and unburned areas that have implications for assumptions used in parameterizing models of active layer and near-surface permafrost dynamics and hydrologic processes.
2. Establish a foundation for integration of field measurements with cryohydrogeologic numerical models through relations between soil physical and soil hydraulic properties that establish the magnitudes and ranges of properties in the boreal forest biome in interior Alaska.

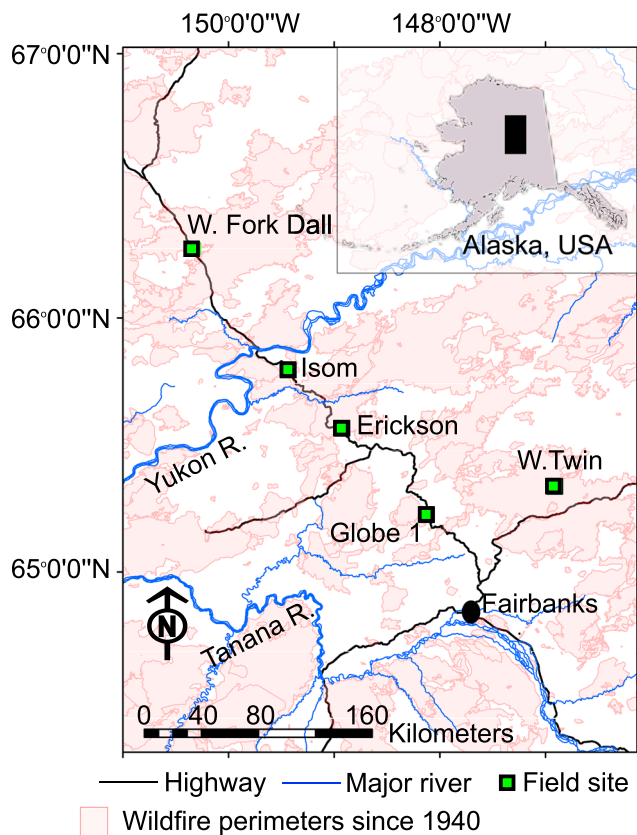


Figure 1. Location map of the field sites in Alaska USA. The black box within the state of Alaska marks the approximate extent of the smaller scale map. Map projection is WGS84. Plotted wildfire perimeters are those that burned since 1940 and were obtained from the Alaska Interagency Coordination Center, <https://fire.ak.blm.gov/predsvcs/maps.php> (accessed 3 August 2017).

September 2016 (Table 1) when active layer thicknesses tend to be greatest to facilitate sampling deeper mineral soils. One soil pit was dug at each sampling location. The thickness of the organic layer was measured by summing the thicknesses of the Oi (slightly decomposed), Oe (moderately decomposed), and Oa (highly decomposed) horizons (excluding buried O horizons at depth) into a total thickness, following the convention of Johnson et al. (2013). This total thickness is referred to as the organic layer thickness herein. Depth to permafrost was assessed using frost probing with a 1.25-m probe, assuming the September 2016 sampling represents the approximate maximum thaw depth and frost probing depths were assumed to be the active layer thickness. At some locations, frost probing was done at the bottom of the soil pit extending the depth of active layer thickness detection. Root depths and diameters were noted during soil pit excavation. Mineral soil samples ($N=4$ per sampling location) were collected from one soil pit at each site at the depths in Table 1 by excavating a bench, then driving in the coring device perpendicular to the ground surface, and then excavating the intact core. Two depths were sampled at the WFD B and WT B sites; these are followed by the “deep” descriptor after the sample name in Table 1. At the G1 B and all WT sites, the presence of stones in the soil required multiple coring attempts to retrieve an intact sample. The shorter cores from the field plots for field saturated hydraulic conductivity (K_{fs}), soil-water retention, soil particle size, loss on ignition (LOI) for determination of organic matter content, and dry ρ_b were 4.8-cm inner diameter and 6 cm long. The presence of roots or root holes was also noted in these 6-cm cores. Some longer cores for soil-water retention analysis were 4.8-cm inner diameter and 13 cm long; these cores were subsampled during analysis. At a limited number of sites, Rhodamine dye tracer was applied by ponding water at the interface of the organic layer and the mineral soil, and a trench was excavated downslope to observe active preferential flow paths. For the site names (Table 1)

3. Synthesize observed physical, hydraulic, and thermal soil properties to develop testable hypotheses related to the influence of subsurface characteristics on coupled water and energy transfer processes in boreal Alaska.

2. Methods

2.1. Field Site Selection

The field sites are located in interior Alaska, primarily along the Dalton Highway and Steese Highway (Figure 1) in boreal forests within the zone of I discontinuous permafrost. These sites span a broadly mapped gradient of permafrost conditions with increasingly discontinuous permafrost expected moving southward (Jorgenson et al., 2008). Mean annual air temperature ranges from < -5 to -2 °C (Walvoord et al., 2019). The soil landscape classifications are either silty or rocky uplands in the Jorgenson et al. (2013) framework. Wildfire is a recent disturbance at some of the sites (Table 1). The dominant tree species at the sites is black spruce (*Picea mariana* (Mill.) Britton, Sterns & Poggenb.), which is the most common tree species in interior Alaska boreal forests (Osterkamp et al., 2000; Van Cleve & Viereck, 1983). The West Twin sites have white spruce present (*Picea glauca* (Moench) Voss; Kelsey et al., 2012). All physical and hydraulic property data are publicly available (Ebel, 2018): <https://www.sciencebase.gov/catalog/item/59974466e4b0b589267bbbc8>. All thermal property data are publicly available (Ebel, 2019a): <https://www.sciencebase.gov/catalog/item/5bf32440e4b045bfcae0c37c>.

2.2. Sample Collection

Sample locations at individual sites and depths of sampling were selected to balance characterization of soil layers with co-location with ground based geophysical surveys including shallow borehole nuclear magnetic resonance and surface direct current electrical resistivity (Minsley et al., 2018). Samples were collected in late August and early

Table 1
Sampling Locations and Characteristics

Site name	Abbreviation	Rocky/silty catchment-scale classification	Wildfire condition	Wildfire date/name	Date sampled (2016)	Organic layer thickness (cm)	Active layer thickness (cm)	Depth range		UTM	
								6-cm cores (cm)	13-cm cores (cm)	Easting (m)	Northing (m)
West Fork of Dall burned	WFD B	Rocky	Burned	2004 Dall City Fire	2 September	5	>245	35-41		619,005	7,352,251
West Fork of Dall burned deep	WFD BD	Rocky	Burned	2004 Dall City Fire	2 September	5	>245	65-71	70-83	619,005	7,352,251
West Fork of Dall unburned	WFD UB	Rocky	Unburned	NA	2 September	20-25	55	20-26		618,998	7,352,220
West Twin unburned	WT UB	Rocky	Unburned	NA	31 August	20-23	90-100	20-26	35 ^b	503,981	7,246,801
West Twin burned	WT B	Rocky	Burned	2004 Boundary Fire	31 August	9-21	>123	12-22 ^a	20-33	504,105	7,246,786
West Twin burned seep	WT BS	Rocky	Burned	2004 Boundary Fire	31 August	5	>123	10-16	15 ^b	504,050	7,246,690
West Twin burned seep deep	WT BSD	Rocky	Burned	2004 Boundary Fire	31 August	5	>123	35-41		504,050	7,246,690
Globe 1 burned	G1 B	Rocky	Burned	2015 Aggie Creek Fire	30 August	5-7	Rock refusal	5.5-11.5		447,395	7,234,653
Globe 1 toeslope	G1 TS	Rocky	Unburned	NA	30 August	20-30	63	20-26	30 ^b	447,432	7,23,4653
Globe 1 unburned	G1 UB	Rocky	Unburned	NA	30 August	15-20	Not sampled	20-26		447,584	7,234,617
Isom	Isom	Silty	Unburned	NA	3 September	12-13	51	24-30	24-37	388,549	7,299,868
Erickson burned upslope deciduous	E BUD	Silty	Burned	2003 Erickson Creek Fire	4 September	5-8	209	20-26	25-38	411,138	7,273,628
Erickson burned	E B	Silty	Burned	2003 Erickson Creek Fire	4 September	5-14	105	20-26		411,106	7,273,526

Note. Locations are Universal Transverse Mercator (UTM), WGS 84 Zone 6 N, except West Fork of Dall sites, which are Zone 5N. The number of samples, N, is N=4 per site for 6-cm cores and N=1 per site for 13-cm cores. Sites are listed by the rocky and silty catchment classifications and then from north (first) to south (last). Samples denoted as "deep" are from greater depth in the same soil pit.

^aSampled just below variable depth interface between organic layer and mineral soil. ^bSample was taken parallel to the soil surface because of rock or frozen soil obstruction.

for WFD BD and WT BSD, the deep denotes the depth of sampling in the soil profile and is not related to any depth of burning of surface organic layers from the wildfire.

2.3. Field-Saturated Hydraulic Conductivity

Field-saturated hydraulic conductivity (K_{fs}) was measured for the 6-cm cores using a tension permeameter (Mini Disc portable tension infiltrometer, Decagon Devices) at -0.5-cm pressure (nearly saturated). The term field-saturated denotes that conditions were nearly saturated, which is typical of flow conditions in the field because of entrapped air (Reynolds et al., 1983). Measurements of K_{fs} were conducted at “as sampled” volumetric-water contents on cores that were sealed after collection in the field. The permeameter was coupled to the cores in the laboratory using a flexible plastic sleeve and several millimeter of contact sand between the soil sample and the permeameter base. The contact sand enhances hydraulic contact between the permeameter and the sample (Nyman et al., 2010). Samples were placed into a stand with a plastic screen secured to the core base; this facilitates free drainage of water from the base of the core. The lateral confinement of the soil core by the impermeable steel walls allows use of the one-dimensional (vertical) flow equations from Philip (1957, 1969) describing cumulative infiltration with time. Cumulative infiltration with time was analyzed using the differentiated linearization method from Vandervaere et al. (2000) by fitting a linear regression line to the differentiated cumulative infiltration with time against the geometric mean of the square root of time. The differentiated linearization method was chosen because the method is more accurate than the cumulative infiltration methods in the presence of contact sand and can readily identify layered soil conditions that can complicate K_{fs} estimation.

2.4. Soil-Water Retention

The soil-water retention measurements were conducted for the 6-cm cores after the K_{fs} measurements, which necessitated air drying the surface of the soil samples for 48 hr to allow brushing off the contact sand material. Soil samples were then saturated by immersion in water for a period of 14.3 days on average with a standard deviation of 5.8 days. Samples were sequentially wetted up over a period of days by increasing the water depth to be level to the top of the core, with a longer initial wetting period at a shallow water depth to minimize slaking of soil samples (Klute, 1986, p. 643-644). Immersion saturation, followed by measurements during the drying process, produces an estimate of the main drying curve (Klute, 1986). Measurements at 10, 33, 50, and 90 kPa applied pressures were conducted using a 500-kPa pressure plate apparatus (Soilmoisture Equipment 1600F1). The 6-cm core height necessitated longer periods of equilibration in the pressure plate apparatus, on average 9.9 days but over 2 weeks at the higher suctions; equilibration was determined by the cessation of outflow from the pressure plate apparatus. One or two measurements at drier conditions (1 to 2 weeks of air drying) were conducted using a dewpoint potentiometer (Gee et al., 1992) by removing a soil subsample extracted from the core center. After the dewpoint potentiometer measurements were taken, the sample was then oven dried at 60°C for 48 hr to estimate dry ρ_b for conversion of gravimetric to volumetric soil-water content. Sample volumes and corresponding bulk densities were corrected for any shrinkage at the soil-water retention measurement steps by measuring vertical and lateral strain.

For the longer 13-cm cores, the Daniel B Stephens and Associates Soil Testing and Research Laboratory in Albuquerque, NM, estimated soil-water retention using several methods to cover the full range of soil-water retention. These methods include the hanging column (Dane & Hopmans, 2002a), the pressure plate (Dane & Hopmans, 2002b), the dewpoint potentiometer (Gee et al., 1992), and a relative humidity-controlled chamber (Nimmo & Winfield, 2002). For the dewpoint potentiometer and relative humidity-controlled chamber methods, a portion of each sample was removed from the long core. The computer software RETC (van Genuchten et al., 1991) was used to estimate van Genuchten (1980) parameters using the water retention data for both the 6- and 13-cm soil cores. The assumption that $m = 1-1/N$ was used in the van Genuchten (1980) parameter fitting.

2.5. Particle Size

Samples were mechanically dispersed using a mortar and pestle prior to sieving. Measurement of the particle-size distribution was conducted using standard dry-sieving techniques for standard class sizes (Guy, 1969) down to 0.063 mm. Soil aggregates were gently abraded against the sieve surface during sieving. Particle size distributions for the fraction smaller than 0.063 mm (down to 4×10^{-5} mm) were analyzed using

a Coulter LS-230 optical diffraction apparatus (Gee & Or, 2002). Further information on this particle-size determination technique can be found in Winfield (2003).

2.6. Loss on Ignition

LOI measurements to estimate organic matter content (Dean, 1974; Heiri et al., 2001) were conducted at 550°C for 2 hr. The LOI technique, while potentially less accurate than other methods for analyzing the amount of soil organic matter (e.g., O'Donnell et al., 2011), was useful here for comparing relative differences between samples. Soil organic matter is taken here to include plant, animal, and microbial residues at all stages of decomposition in addition to humus and restricted to materials passing through a 2-mm sieve (Nelson & Sommers, 1996).

2.7. Volumetric Specific Heat Capacity and Thermal Conductivity

Subsamples of 13-cm soil cores from six sites (E BUD, G1 TS, Isom, WT B, WT UB, and WFD B) were analyzed for selected soil thermal properties under fully thawed conditions across the full range of water saturations. Soil thermal properties of volumetric specific heat capacity (C) and thermal conductivity (k) were estimated using a KD2 Pro Thermal Properties Analyzer (Decagon Devices, 2016). The KD2 Pro uses a 3-cm dual needle, with one heated needle and one monitoring needle, and the line heat source analysis approximations from Carslaw and Jaeger (1959) and Kluitenberg et al. (1993) with a nonlinear least squares fitting procedure. Thermal properties were measured at approximately 20°C (i.e., fully thawed conditions) from fully saturated to oven-dry conditions with 8 to 11 data points distributed across the range of saturation per sample. All analyses were conducted at the Daniel B. Stephens and Associates Laboratory. Measured thermal properties as a function of saturation were compared against common approximations for C and k typically used in thermal and cryohydrogeologic models (e.g., Atchley et al., 2016; Harp et al., 2015; Langer et al., 2013) in the absence of measured thermal properties: the volume fraction weighted approach for C from Hillel (1980) and the Johansen (1975) approach for k as presented in Peters-Lidard et al. (1998). Values for C for the constituents (i.e., air, water, and solid) are from Table 9.1 in Hillel (1980); organic matter was neglected. The Johansen (1975) approach for k was parameterized with the measured ρ_b , saturated volumetric soil-water content was used to approximate porosity, quartz contents were estimated based upon measured soil texture using Table 2 in Peters-Lidard et al. (1998), and all soils were in the “fine” classification from Farouki (1981, 1986) for approximating the Kersten number, which is based on the percentage by mass of particles $<2 \mu\text{m}$.

2.8. Statistical Analysis

Statistical hypothesis testing for differences in the median values for ‘rocky’ and ‘silty’ and ‘burned’ and ‘unburned’ samples was conducted using the Kruskal-Wallis test (Kruskal & Wallis, 1952) in the Scipy.stats package in Python. Bartlett’s (Snedecor & Cochran, 1983) and Levene’s (Levene, 1960) tests were used to examine the equality of variance for the log-transform of K_{fs} for silty versus rocky and burned versus unburned sites. Assessment of the lognormality of K_{fs} was conducted using the Anderson-Darling test (Anderson & Darling, 1954) in the Statsmodels package in Python. Regression analyses also used the Statsmodels package in Python and used 95% confidence intervals based on a Student’s t distribution. The p -values for regression results were assessed using t tests against the null hypothesis of zero slope. The p -values are classified herein as strong evidence ($p < 10^{-4}$), moderate evidence ($10^{-4} \leq p < 0.05$), suggestive evidence ($0.05 \leq p < 0.1$), and little evidence ($p \geq 0.1$) against the null hypothesis (e.g., Leek & Peng, 2015; Nuzzo, 2014).

3. Results

3.1. Bulk Density (ρ_b)

Silty sites tended to have less variable but greater average ρ_b , between 1.0 and 1.5 g/cm³, whereas rocky sites had a greater variation of average values from 0.57 to 1.61 g/cm³ (Figure 2a and Table 2). There was moderate evidence of differences in ρ_b between rocky and silty catchments ($p=3.81 \times 10^{-2}$) but only suggestive evidence between burned and unburned sites ($p=6.54 \times 10^{-2}$; Tables 2 and 3). The primary driver of ρ_b variation between sampling sites appears to be soil organic matter based on the strong correlations with LOI at rocky ($R^2=0.646$, $p\text{-value} = 7.30 \times 10^{-10}$; Figure 3a) and silty sites ($R^2=0.793$, $p\text{-value} = 2.37 \times 10^{-4}$; Figure 3b), with

Table 2Summary Table of Soil Physical Property and Field-Saturated Hydraulic Conductivity (K_{fs}) Measurements From 6-cm Core Samples

Site name	Abbreviation	Rocky/silty catchment-scale classification	Depth range (cm)	ρ_b	LOI	Soil-particle size			θ_s	K_{fs}
				Mean (g/cm ³)	Mean (%)	Sand (%)	Silt (%)	Clay (%)	Mean (cm ³ /cm ³)	Geometric mean (m/s)
West Fork of Dall burned	WFD B	Rocky	35-41	1.10	13.0	34.0	29.3	36.6	0.66	1.8E-05
West Fork of Dall burned deep	WFD BD	Rocky	65-71	1.08	13.5	40.7	29.0	30.3	0.62	6.1E-07
West Fork of Dall unburned	WFD UB	Rocky	20-26	1.61	12.6	29.7	41.9	28.4	0.65	4.0E-07
West Twin unburned	WT UB	Rocky	20-26	1.61	6.9	57.1	22.4	20.5	0.47	2.7E-07
West Twin burned	WT B	Rocky	12-22 ^a	0.57	29.6	57.5	27.2	15.3	0.83	2.7E-05
West Twin burned seep	WT BS	Rocky	10-16	0.85	19.1	57.3	26.2	16.5	0.71	1.4E-07
West Twin burned seep deep	WT BSD	Rocky	35-41	1.00	17.5	39.6	36.6	23.9	0.67	8.8E-07
Globe 1 burned	G1 B	Rocky	5.5-11.5	0.69	17.5	73.5	16.8	9.7	0.64	1.8E-03
Globe 1 toeslope	G1 TS	Rocky	20-26	1.18	7.6	30.9	46.1	23.0	0.63	1.3E-06
Globe 1 unburned	G1 UB	Rocky	20-26	0.68	20.8	41.8	35.2	23.0	0.81	4.9E-05
Isom	Isom	Silty	24-30	1.06	11.4	18.9	61.4	19.7	0.69	3.7E-06
Erickson burned upslope deciduous	E BUD	Silty	20-26	1.28	5.9	18.1	53.6	28.4	0.60	2.5E-05
Erickson burned	E B	Silty	20-26	1.50	6.0	17.3	48.6	34.0	0.56	2.3E-07

Note. The number of samples (N) for the 6-cm cores at each site is 4. The dry bulk density is denoted as ρ_b . Loss on ignition is denoted as LOI. The mean is the arithmetic mean. Saturated soil-water content is θ_s . Sand refers to particles from 2 to 0.063 mm. Silt refers to particles from 0.063 to 0.002 mm. Clay refers to particles <0.002 mm. Sites are listed by the rocky and silty catchment classifications and then from north (first) to south (last). Samples denoted as "deep" are from greater depth in the same soil pit.

^aSampled just below variable depth interface between organic layer and mineral soil.

ρ_b declining exponentially with increasing soil organic matter (Table 2). The majority of the data lie within the 95% confidence intervals of the regression (Figures 3a and 3b).

3.2. LOI

Rocky sites had greater LOI values (site averages ranged from 6.9 to 29.6% by mass) than silty sites (site averages ranged from 5.9 to 11.4%; Table 2). Burned sites tended to have smaller site averaged LOI values compared to unburned sites, especially at rocky sites (Figure 2b and Table 2). LOI was also more variable at rocky sites, in particular at the WT and G1 sites (Figure 2b). LOI had moderate evidence of differences between rocky and silty ($p=1.62 \times 10^{-4}$) catchment sites but only suggestive evidence of differences between burned and unburned sites ($p=8.13 \times 10^{-2}$; Table 3). Silty sites are significantly lower in soil organic matter, based on LOI, compared to rocky sampling sites (Table 3).

3.3. Saturated Soil-Water Content (θ_s)

Average θ_s at all sites is relatively large, reflecting fine-grained soils, substantial organic matter content, and moderate to low bulk densities (Figure 2c and Table 2). No clear trend was present in average θ_s for rocky and silty or burned and unburned sites (Table 2). θ_s had little evidence of differences between rocky and silty ($p=1.05 \times 10^{-1}$) and little evidence of differences between burned and unburned catchment sites ($p=7.21 \times 10^{-1}$; Table 3). Correlations for θ_s are linear relative to soil organic matter (regressed against LOI) with a relatively high $R^2=0.58$ ($p=1.21 \times 10^{-10}$; Figure 3c) and inverse and linear relative to ρ_b also with a relatively high $R^2=0.62$ ($p=1.44 \times 10^{-11}$; Figure 3d). Note that because θ_s only had little evidence of differences between rocky versus silty sites, the data were combined for the regressions in Figures 3c and 3d. The 95% confidence intervals of the regression generally encapsulate the data (Figures 3c and 3d).

3.4. Field-Saturated Hydraulic Conductivity (K_{fs})

The differences in geometric mean K_{fs} between sites span 4 orders of magnitude from approximately 10^{-3} to 10^{-7} m/s¹ (Table 2). No clear distinction in geometric mean K_{fs} between rocky and silty or burned and unburned sites was present (Table 2 and Figure 2d). K_{fs} had little evidence of differences between rocky and silty ($p=9.10 \times 10^{-1}$) or burned versus unburned sites ($p=2.47 \times 10^{-1}$; Table 3) even though some burned sampling sites have larger K_{fs} (Figure 2d and Table 2). The burned site K_{fs} values, relative to unburned sites,

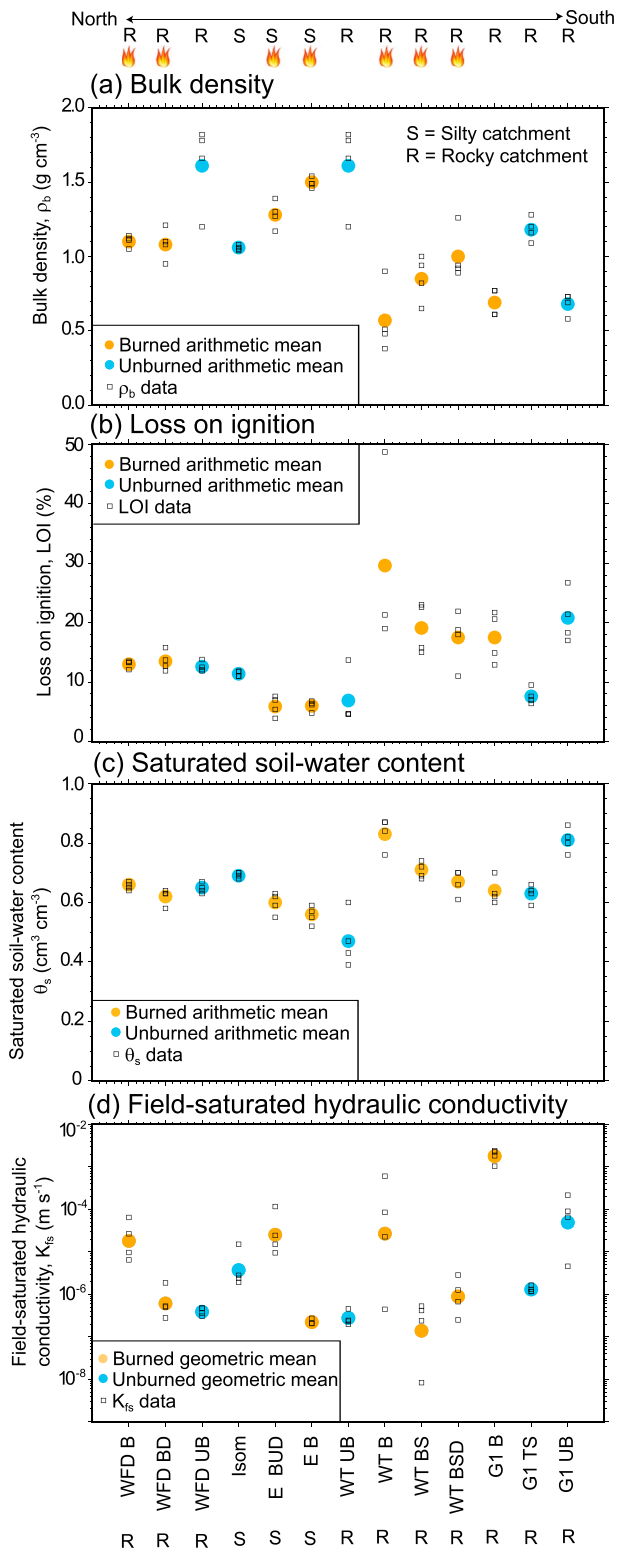


Figure 2. Site data and mean values for (a) bulk density, (b) loss on ignition, (c) saturated soil-water content, and (d) field-saturated hydraulic conductivity.

had moderate evidence of higher variability in magnitude based on both the Bartlett's ($p=3.32 \times 10^{-2}$) and Levene's ($p=1.17 \times 10^{-2}$) tests on log-transformed K_{fs} . The variance in $\log(K_{fs})$ for rocky compared to silty sites showed little evidence of significant differences (Bartlett's test, $p=1.53 \times 10^{-1}$; Levene's test $p=3.05 \times 10^{-1}$). The log-transformed K_{fs} with all sites taken together did not fit a normal distribution, based on the Anderson-Darling test (test statistic=1.469, $p=7.6 \times 10^{-4}$), indicating K_{fs} is not lognormally distributed. K_{fs} was not well correlated with the soil physical properties. The strongest correlation for K_{fs} was against ρ_b (an exponential relation) with a moderately low R^2 ($R^2=0.36$, $p=3.92 \times 10^{-6}$; Figure 3e). Note that because K_{fs} showed little evidence of differences between rocky versus silty sites (Table 3), the data were combined for the regression in Figure 3e. Data were generally contained within the 95% confidence intervals of the regression (Figure 3e).

3.5. Particle-Size Distribution

The particle-size distribution closely followed the rocky and silty catchment classifications (Table 2). The Isom sites were classified as silt loams, and the E B and E BUD sampling sites were classified as silty clay loams (Figure 4 and Table 2). The rocky catchment sites spanned a much broader range in particle-size distribution, from the clay loam sites of WFD to the sand and sandy loam sites at G1 B (Figure 4 and Table 2). The sand fraction had strong evidence of differences between rocky and silty ($p=5.06 \times 10^{-7}$) but little evidence of differences between burned versus unburned catchment sites ($p=2.55 \times 10^{-1}$; Table 3). The silt fraction also had strong evidence of differences between rocky and silty ($p=6.44 \times 10^{-7}$) and moderate evidence of differences between burned and unburned catchment sites ($p=4.13 \times 10^{-2}$; Table 3). Sand fractions were greater at rocky catchments sites (Table 2). Clay had little evidence of differences for any of the classifications, and gravel had moderate evidence of differences for the silty versus rocky classification ($p=1.28 \times 10^{-2}$; Table 3). Linear regressions for θ_s against sand ($R^2=0.031$) and silt ($R^2=0.004$), showed little evidence for linear relations based on t tests, but the regression had moderate evidence for linear relations for clay ($R^2=0.089$, $p=3.50 \times 10^{-2}$), suggesting that particle-size distribution was not a strong predictor of θ_s . Linear regressions for log-transformed K_{fs} had very low R^2 values, and the regression had little evidence for a linear relation for silt ($R^2=0.05$, $p=0.117$); the regressions for sand ($R^2=0.11$, $p=0.021$, positive slope) and clay had moderate evidence for linear relations ($R^2=0.14$, $p=0.008$, negative slope), but the low R^2 values suggest poor predictive capability. Because θ_s and K_{fs} showed little evidence against the null hypothesis for differences between rocky versus silty sites, the aforementioned regressions were for all the data combined.

3.6. Soil-Water Retention

Silty catchment sites tended to have less variability in soil-water retention characteristics, relative to rocky sites. Correspondingly, there was less variation in site-averaged van Genuchten parameters α and N at silty sites compared to rocky sites (Table 4). There was moderate evidence of differences in the van Genuchten parameters α ($p=1.10 \times 10^{-2}$) and N ($p=2.94 \times 10^{-2}$) between rocky and silty classifications but little evidence of differences in α or N between burned and unburned

Table 3

Statistical Tests for Soil Physical and Hydraulic Properties Between Catchment Classifications of Rocky ($N=39$) Versus Silty ($N=11$) and Sampling Site Classifications of Burned ($N=30$) and Unburned ($N=20$) Using the Kruskal-Wallis Test (Kruskal & Wallis, 1952) in the Scipy.stats Package in Python

Soil property	Rocky versus silty	Burned versus unburned
	p -value (—)	p -value (—)
Bulk density (ρ_b)	3.81E–02	6.54E–02
Loss on ignition (LOI)	1.62E–04	8.13E–02
Saturated soil-water content (θ_s)	1.05E–01	7.21E–01
Field-saturated hydraulic conductivity (K_{fs})	9.10E–01	2.47E–01
Sand fraction	5.06E–07	2.55E–01
Silt fraction	6.44E–07	4.13E–02
Clay fraction	1.71E–01	3.47E–01
Gravel fraction	1.28E–02	7.74E–01
van Genuchten alpha	1.10E–02	5.73E–01
van Genuchten N	2.94E–02	3.10E–01

Note. Bolded values have $p < 0.05$.

classifications (Table 3). A major difference between the soil-water retention curves appears to be the θ_s values (Figures 5 and 2c), potentially caused by differences in soil organic matter (Table 2). The soil-water retention curves for the longer 13-cm cores tend to decrease more steeply around -10^3 cm of pressure head, compared to the curves for the 6-cm cores (Figure 5), which may be the result of the absence of soil-water retention data in this pressure head region in the 6-cm cores because of air compressor issues. Neither the 6-cm nor the 13-cm cores showed substantial differences in soil-water content between saturation and 100 cm of pressured head at most sites, with the exception of the rocky G1 sites (Figure 5). Regressions for van Genuchten parameters were separated into rocky and silty classifications because of the significant differences (mentioned above). Van Genuchten parameters alpha and N were not well correlated with soil physical properties (Figures 6a and 6d). The strongest relations against physical properties, based on R^2 , were with ρ_b for rocky sites (alpha has $R^2=0.319$ and $p=1.75 \times 10^{-4}$, N has $R^2=0.184$ and $p=6.38 \times 10^{-3}$; Figures 6a and 6d) and LOI (alpha has $R^2=0.127$ and $p=2.57 \times 10^{-2}$, N has $R^2=0.177$ and $p=7.66 \times 10^{-3}$; not shown in Figure 6). Regression relations for the van Genuchten parameters alpha and N were strongest (based on R^2 and p -values) for soil-particle size fractions at rocky sites, specifically sand (alpha has $R^2=0.313$ and $p=2.12 \times 10^{-4}$, N has $R^2=0.26$ and $p=9.19 \times 10^{-4}$; Figures 6b and 6e) and clay percentages (alpha has $R^2=0.528$ and $p=1.61 \times 10^{-7}$, N has $R^2=0.490$ and $p=7.15 \times 10^{-7}$; Figures 6c and 6f). For the silty sites there was little evidence to support correlations between the van Genuchten parameters alpha and N and either ρ_b , LOI, or soil-particle size fractions.

3.7. Volumetric Specific Heat Capacity (C) and Thermal Conductivity (k)

Measured soil thermal properties of C and k were not appreciably different at the rocky and silty sites despite substantial differences at sites for saturated soil-water content and dry bulk density (Table 5 and Figures 7a–7d). The measured C values generally increased linearly with saturation, but the WT UB, WFD BD, Isom, and E BUD C values increased nonlinearly at near-saturated conditions between saturations of 0.9 to 1 (Figures 7a and 7b). Measured values of k for the WT UB, WFD BD, Isom, and E BUD also increased nonlinearly at near-saturated conditions between saturations of 0.9 to 1 (Figures 7c and 7d). Estimated values of C from the Hillel (1980) approach matched measured values well, based on root-mean-squared error between estimated and measured (Table 5), for both rocky and silty sites, with the exception of the G1 TS site. Estimated values for C at the rocky G1 TS site had a large root-mean-squared error of nearly 0.6, and the error is concentrated at saturations from 0.1 to 0.5 (Figure 7a and Table 5). Estimated values of k matched measured values well for silty sites with small root-mean-squared error (Table 5), although the Johansen (1975) approach did not capture the nonlinear k increases between saturations of 0.9 to 1 at Isom and E BUD (Figure 7d) nor can the approach estimate k at saturation ≤ 0.1 for fine-textured soils (Peters-Lidard et al., 1998). Estimated values for k for rocky sites were of mixed quality. The WT B and WFD BD sites were estimated well by the Johansen (1975) approach with small root-mean-squared errors (Table 5) although the estimated values do not capture the k increase at saturations from 0.9 to 1 at WFD BD (Figure 7c). Estimated values of k at the rocky sites G1 TS and WT UB were poor across all saturations with large root-mean-squared errors (Table 5). Greater ρ_b was the most obvious difference in Table 5 between the G1 TS and WT UB sites and the other thermal property sites.

3.8. Root Characteristics in Mineral Soil

Burned sites tended to have roots in the soil profile that penetrated to greater depth, relative to unburned sites (Table 6). There were no obvious differences between silty versus rocky sites for root diameters and depths. Sites with obvious frozen soil layers, interpreted to be permafrost, such as the Isom and WFD UB locations had roots concentrated just below the organic layer interface with the mineral soil, along with a few small roots (≤ 0.5 -mm diameter) and an occasional larger root (1-mm diameter) that stopped at the permafrost interface. Burned sites, most notably the E B, E BUD, and WFD B sites, had roots concentrated in the top 0.5 m below the organic layer interface with the soil but also had small to moderate size roots

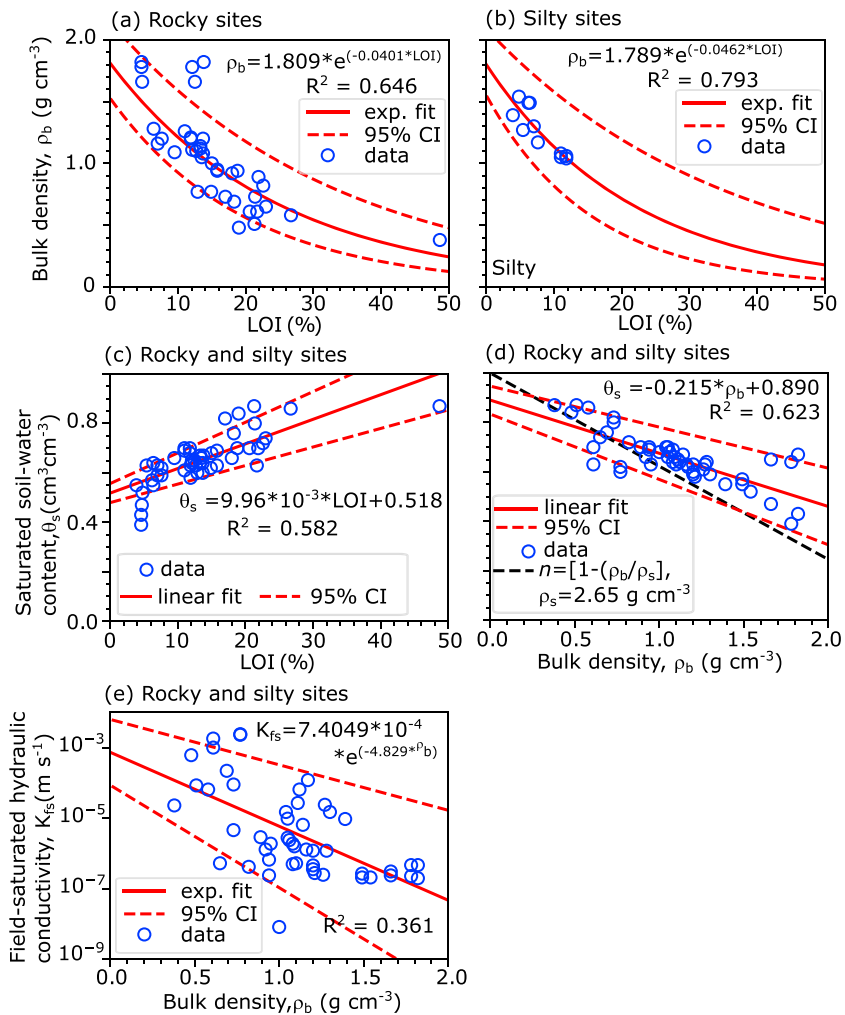


Figure 3. (a–e) Regressions and 95% confidence intervals for selected soil physical and hydraulic properties. Parts (a) and (b) are separated into rocky versus silty sites because of moderate evidence of differences between rocky and silty sites for bulk density and loss on ignition.

(0.5- to 1-mm diameter) down to the base of the soil pit >1-m depth below the surface (Table 6). Observations of roots in the 6-cm soil cores confirmed the observations from the soil pits with clear root presences at E B, E BUD, and WFD B sites and roots up to 1-mm diameter. The 6-cm long soil cores also had notable root presence for some cores at the G1 B, WT B, and WT UB sites with roots up to 1-mm diameter.

4. Discussion

4.1. Silty and Rocky, Burned and Unburned Classifications

Differences in soil-particle size distribution and physical properties (LOI, ρ_b) between rocky and silty soils extends beyond previous determinations and results in unexpectedly large variations in soil-water retention (Figure 5) and van Genuchten parameters (Table 4). The silty versus rocky catchment classification for boreal forest of interior Alaska from Jorgenson et al. (2013) proved useful in separating upland soil landscapes by soil physical and hydraulic properties. The largest differences between rocky and silty sites were, not surprisingly, in soil-particle size distribution (Figure 4 and Table 2) because that was the original basis for the classification. Differences in soil organic matter (LOI) and ρ_b (Figures 2a and 2b and Table 2) were less expected. The combination of differences in soil-particle size distribution and physical properties (LOI, ρ_b) led to variations in soil-water retention (Figure 5) and corresponding van Genuchten parameters (Table 45). The van Genuchten parameters alpha and N that can be used to represent soil-water retention

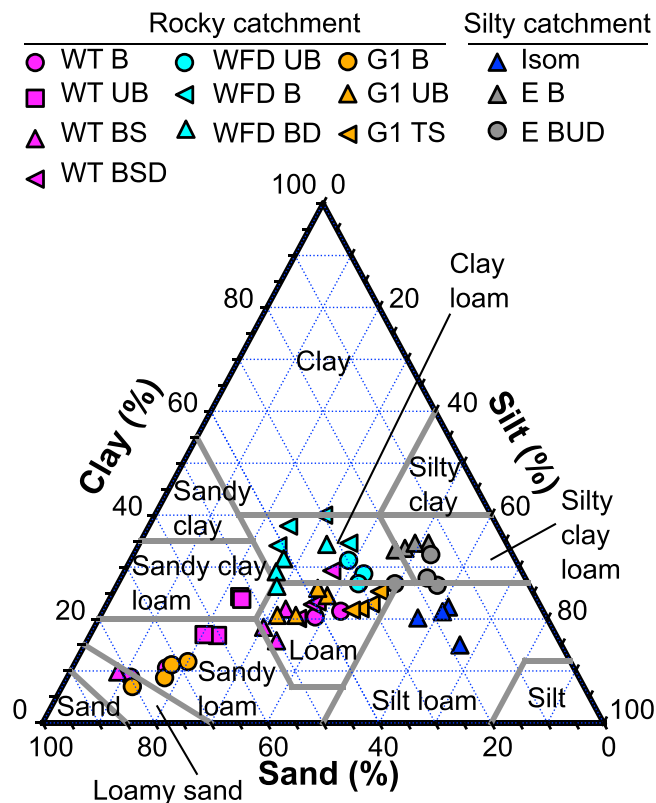


Figure 4. Sand-silt-clay percentages for the study sites overlain on the U.S. Department of Agriculture soil-textural triangle.

relations were more strongly correlated with soil-particle size distribution, specifically clay and sand fractions, than ρ_b and soil organic matter (Figures 6a-6f). This agrees with prior work showing the dominant influence of soil texture in determining soil-water retention (Carsel & Parrish, 1988; Saxton et al., 1986; Zhuang et al., 2001). The correlation between soil organic matter and θ_s , however, leads to an indirect influence on soil-water retention. Between-site differences in soil organic matter were important for determining both ρ_b and θ_s (Figures 3a-3d). The linearly increasing trend between θ_s and soil organic matter (Figure 3c) has been shown for sites outside boreal forest environments where soil organic matter is relatively high, such as in the meadow study in the Sierra Nevada mountains, CA, by Ankenbauer and Loheide (2017). Prior work to develop relations between soil organic matter and soil hydraulic properties showed similar results, in terms of soil organic matter increasing θ_s , but covered a smaller range in soil organic matter from 0 to 8% (Rawls et al., 2003) or less than 10% (Saxton & Rawls, 2006). The relations for θ_s also show substantial deviations from the commonly used estimates of θ_s (i.e., approximately porosity, n) from ρ_b , such as the relation $n = (1 - [\rho_b/\rho_s])$ with solid particle density, ρ_s , assumed to be 2.65 g/cm³ in soils (see Figure 3d). Field conditions of θ_s may be slightly lower, because of entrapped air, than the laboratory-saturated values of θ_s reported here. When ρ_s is altered from purely mineral soil to also have substantial organic contributions, the θ_s to ρ_b relation has a gentler slope with higher θ_s at large ρ_b and lower θ_s at small ρ_b (Figure 3d). The data presented here cover the high soil organic matter contents that are typical of boreal forest soils in interior Alaska, which are outside the range of relations from agricultural soils that are reported in the literature. Conclusions here regarding soil organic matter influences on soil hydraulic properties are restricted to

particulate organic matter in mineral soil and do not apply to Oi, Oe, and Oa horizons that are known to have distinct soil hydraulic properties from mineral soil (e.g., Carey & Woo, 2001; Hinzman et al., 1991; Quinton & Marsh, 1999).

Table 4
Unsaturated Flow Model Parameters for the van Genuchten (1980) Approach for Estimating Soil-Water Retention

Site name	Abbreviation	Rocky/silty catchment-scale classification	Depth range (cm)	θ_s (cm ³ /cm ³)	θ_r (cm ³ /cm ³)	α (cm ⁻¹)	N_{VG} (—)	R^2 (—)
West Fork of Dall burned	WFD B	Rocky	35-41	0.66	0	2.77E-03	1.327	0.97
West Fork of Dall burned deep	WFD BD	Rocky	65-71	0.62	0	4.66E-03	1.286	0.97
West Fork of Dall unburned	WFD UB	Rocky	20-26	0.65	0	7.17E-03	1.255	0.97
West Twin unburned	WT UB	Rocky	20-26	0.47	0	1.01E-02	1.237	0.97
West Twin burned	WT B	Rocky	12-22 ^a	0.83	0	1.87E-02	1.214	0.97
West Twin burned seep	WT BS	Rocky	10-16	0.71	0	1.88E-02	1.186	0.96
West Twin burned seep deep	WT BSD	Rocky	35-41	0.67	0	1.81E-02	1.204	0.98
Globe 1 burned	G1 B	Rocky	5.5-11.5	0.64	0	2.93E-01	1.190	0.98
Globe 1 toeslope	G1 TS	Rocky	20-26	0.63	0	1.67E-02	1.269	0.96
Globe 1 unburned	G1 UB	Rocky	20-26	0.81	0	8.81E-02	1.219	0.95
Isom	Isom	Silty	24-30	0.69	0	6.74E-03	1.276	0.98
Erickson burned upslope deciduous	E BUD	Silty	20-26	0.60	0	7.87E-03	1.275	0.97
Erickson burned	E B	Silty	20-26	0.56	0	4.89E-03	1.260	0.97

Note. All parameters were estimated using the RETC software (van Genuchten et al., 1991) fit to the measured soil-water retention data. The number of samples (N) for the 6-cm cores at each site is 4. Sites are listed by the rocky and silty catchment classifications and then from north (first) to south (last). Samples denoted as "deep" are from greater depth in the same soil pit.

R^2 is the coefficient of determination comparing the model fit to the measurements.

θ_s is saturated soil-water content, θ_r is residual soil-water content, α and N_{VG} are parameters in the van Genuchten model.

^aSampled just below variable depth interface between organic layer and mineral soil.

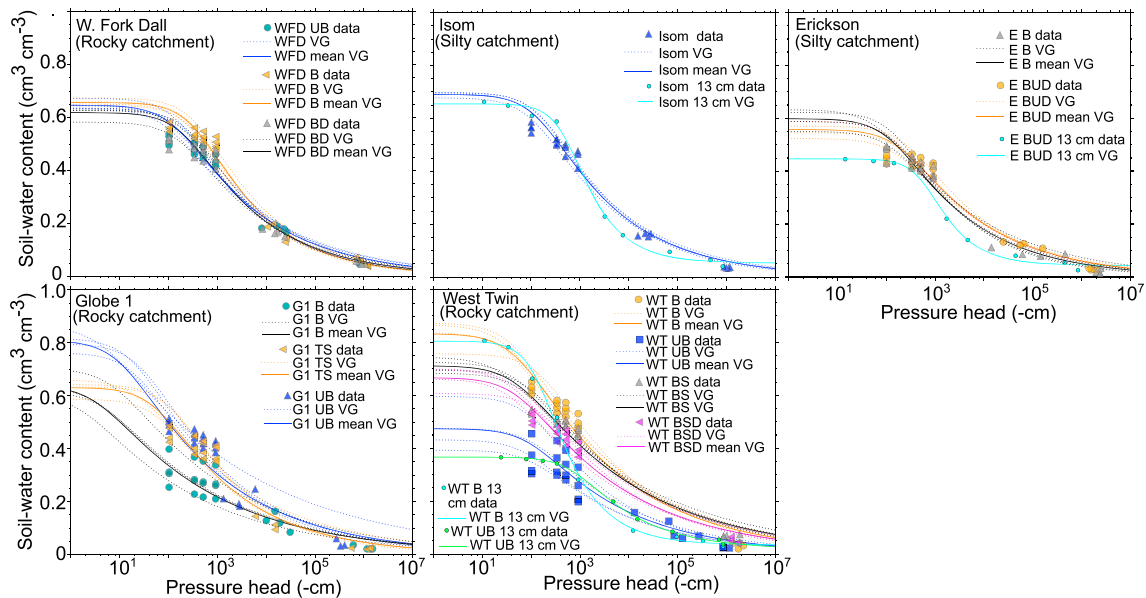


Figure 5. Soil-water retention measurements and estimated functional relations based on the van Genuchten (1980) approach. Functional relation fitting was done using the RETC software (van Genuchten et al., 1991). Van Genuchten (VG) summary parameters are provided in Table 4. Note that θ_s values are not plotted because of the log scale, but the measured θ_s values were used in the RETC curve fitting. All data are from 6-cm cores unless labeled in legend as 13-cm cores.

Overall, the soil thermal properties showed minor differences between rocky and silty sites despite substantial differences in bulk density. The methods for estimating C and k from Hillel (1980) and Johansen (1975), respectively, agreed moderately well with the measured data, with the exception of C at G1 TS between saturation of 0.1 to 0.5 and for k at the G1 TS and WT UB sites. The greater ρ_b at the G1 TS and WT UB sites

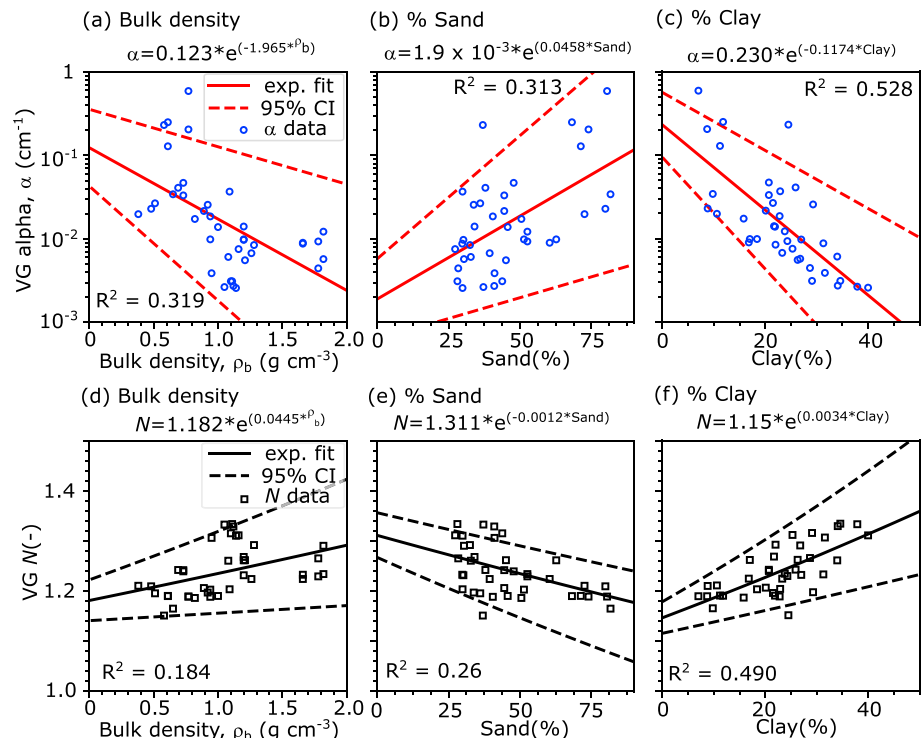


Figure 6. Regression relations at rocky field sites for the van Genuchten (VG) parameters (a-c) α and (d-f) N against selected soil physical properties.

Table 5
Sampling Locations for Soil Subsamples From 13-cm-Long Soil Cores

Site name	Abbreviation	Rocky/silty catchment-scale classification	Wildfire condition	Depth range (cm)	Bulk density (g/cm^3)	Saturated soil-water content (cm^3/cm^3)	Root mean squared error	
							Volumetric specific heat capacity ($\text{MJ} \cdot \text{m}^{-3} \cdot \text{K}^{-1}$)	Thermal conductivity ($\text{W} \cdot \text{m}^{-1} \cdot \text{K}^{-1}$)
West Fork of Dall burned deep	WFD BD	Rocky	Burned	70-83	1.26	0.526	0.232	0.137
West Twin unburned	WT UB	Rocky	Unburned	35 ^a	1.8	0.372	0.259	0.557
West Twin burned	WT B	Rocky	Burned	20-33	0.45	0.81	0.159	0.075
Globe 1 toeslope	G1 TS	Rocky	Unburned	30 ^a	1.6	0.417	0.599	0.430
Isom	Isom	Silty	Unburned	24-37	1.03	0.672	0.298	0.144
Erickson burned upslope deciduous	E BUD	Silty	Burned	25-38	1.52	0.449	0.312	0.178

Note. See Table 1 for further details. Root-mean-square error is between measured data and estimated values from the Hillel (1980) approach for volumetric specific heat capacity and the Johansen (1975) approach for thermal conductivity.

^aSample was taken parallel to the soil surface because of rock or frozen soil obstruction.

is likely the result of higher gravel (i.e., particles larger than 2 mm) percentages by mass. The WT UB site average was 19.9% gravel by mass with a standard deviation of 13.9%. The G1 TS site did not have cores where soil-particle size distribution was measured at the same depth as the 13-cm core used for thermal properties. Field notes indicated a gravel layer at 28-cm depth exactly where the 13-cm core at G1 TS was taken. The nearby G1 B site had 34.4% gravel by mass with a standard deviation of 7.1%. The soil particle size distribution data for the other sites indicate that 82% of samples had no gravel and only one sample (at the WT BSD site) had >1% gravel by mass. The gravel content at WT UB and G1 TS also may present issues for the KD2 Pro Thermal Properties Analyzer because of inconsistent thermal contact with the sample between measurements at different saturations that are not at exactly the same location in the sample, for example, the scatter in the G1 TS measurements for C . These measurements also highlight that estimates of mineral soil k based largely on ρ_b (e.g., using the method from Peters-Lidard et al., 1998) may lead to overestimating k by up to $0.56 \text{ W} \cdot \text{m}^{-1} \cdot \text{deg} \cdot \text{K}^{-1}$ when substantial gravel fractions are present.

The impact of gravel content of mineral soils on thermal and hydrologic model parameterizations shown here for interior Alaska has also been highlighted in other areas where permafrost is present. For example, studies focusing on the Qinghai-Tibetan Plateau have showed that thermal conductivity increases with increasing gravel content in mineral soils (Pan et al., 2017; Yi et al., 2013, 2018). Including the effects of gravel on thermal properties, compared to default soil parameters in land surface models for the Qinghai-Tibetan Plateau, reduced model errors of active layer depth by 74 to 84 % and errors of simulated basal permafrost boundary depth by 35 or 34 % (Yi et al., 2018). Pan et al. (2017) showed that inclusion of gravel impacts on thermal conductivity reduced mean model bias by 25 and 48.3% for sites on the Qinghai-Tibetan Plateau region. Both Pan et al. (2017) and Yi et al. (2018) prescribed regional data collection campaigns establishing relations for thermal and hydraulic properties to improve modeling of freeze-thaw processes and permafrost dynamics. The data summarized in this work address the recommendations by Pan et al. (2017) and Yi et al. (2018) for interior Alaska.

By contrast, burned and unburned conditions at sampling sites had little impact on physical, hydraulic, and thermal properties of mineral soil at the depths of this investigation. Wildfires in interior Alaska do not typically have the major impacts on physical and hydraulic properties of mineral soil (e.g., Brown et al., 1983; Viereck & Dyrness, 1979) that can be present in the continental United States, for example, see reviews in Certini (2005), Ebel and Moody (2017), and Ebel (2012, 2019b). The direct heat impulse into mineral soil during interior Alaska wildfire is minimal to absent owing to the thick, often moist, organic layer at the surface (Brown et al., 1983; Dyrness & Norum, 1983; Viereck & Dyrness, 1979), which reduces or eliminates direct wildfire impacts on mineral soil properties. This is likely why no substantial fire impacts were observed in this work, given that the average sampling depth was nearly 27 cm below land surface, sometimes beneath a thick organic layer (Table 1). The relatively small spatial scale of the samples (109 cm^3

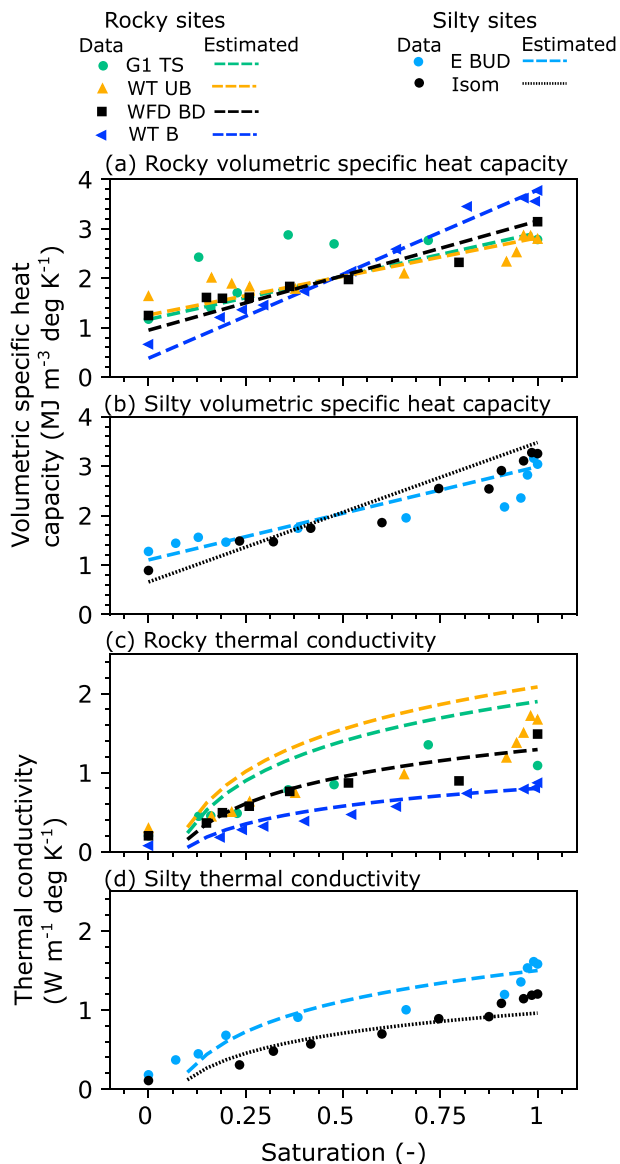


Figure 7. Measured data and estimated volumetric specific heat capacity and thermal conductivity for (a and c) rocky and (b and d) silty sites. Estimated values for volumetric specific heat capacity are based on the approach from Hillel (1980) and for thermal conductivity from Johansen (1975) as summarized in Peters-Lidard et al. (1998). Root-mean-squared error between measured and estimated soil thermal properties are given in Table 5.

for the 6-cm-long cores) also likely precludes sampling large soil pipes or conduits from burned root holes or other subsurface features that can serve as important flow paths in permafrost landscapes (e.g., Carey & Woo, 2000; Carey & Woo, 2002; Koch, Ewing, et al., 2013).

4.2. Preferential Flow

One exception to minimal wildfire impacts on soil hydraulic properties observed at the study sites were differences in K_{fs} variability, which showed moderate evidence of being greater at burned sites. Some of the soil cores at burned sites had 0.5- to 1-mm roots, most notably the E BUD, G1 B, WT B, and WFD B sites (Table 6). These sites also had some of the highest K_{fs} values (Figure 2d). Root holes often serve as paths for preferential flow (e.g., Ghestem et al., 2011; Johnson & Lehmann, 2006; Noguchi et al., 1999), with preferential flow defined by Nimmo (2012) as flow through a preferred portion of a medium through pathways, pores, or macroscopic subvolumes with high speed and volumetric contributions that can bypass the bulk of the soil matrix even in unsaturated conditions. The roots observed at these burned sites penetrated the entire soil column (in the soil pit) to depths of 1.2 m at WFD B and 1.05 m at E BUD. Root systems in boreal forests have roots of primarily black and white spruce concentrated in the top 20 cm (Ruess et al., 2006; Tryon & Chapin, 1983) because shallow permafrost typically prevents deep drainage (Jorgenson et al., 2010), soil temperatures below 6°C reduce root growth (Alvarez-Uria & Körner, 2007; Körner & Hoch, 2006; Tryon & Chapin, 1983), and root growth is minimal below 1°C (Bell & Bliss, 1978; Billings et al., 1977; Chapin, 1974; Kummerow & Ellis, 1984). The deeper root growth at the E BUD and WFD B sites may reflect deep permafrost (beyond the frost probing depth of $>1.25\text{ m}$). At the E BUD site, vegetation prior to and after wildfire was primarily deciduous. At the WFD B site, postfire vegetation consisted of grasses and dwarf birch (*Betula nana* L.). The dye tracer tests at the WFD B site showed clear evidence of preferential flow at the interface between the organic layer and mineral soil and also along root holes within the mineral soil (Figure 8).

A shift from coniferous-tree dominated to deciduous shrub-dominated communities after wildfire has been observed in boreal regions (Beck et al., 2011; Johnstone et al., 2010; Johnstone & Chapin, 2006; Tape et al., 2006) and is expected to continue in the 21st century (Pastick et al., 2017). Deciduous annual roots tend to penetrate deeper, relative to black spruce root systems (Tryon & Chapin, 1983). The K_{fs} results reported here are congruent with the conceptual model that deeper active layers and deciduous vegetation promote deeper root growth that enhance vertical preferential flow and increase K_{fs} , potentially moving water rapidly to depth and enhancing drainage of burned upland land-

scapes. The dye tracer tests provide visual support for preferential flow through root holes in mineral soil at the WFD B site (Figure 8), though the extent to which preferential flow enhances permafrost thaw at this site remains unquantified. The conceptual model of fire-vegetation-soil property interaction, formulated by the results of this study, is one that requires additional testing through both model and field investigations to evaluate for robustness and ubiquity.

The within-site variability in soil-water retention could potentially provide preferential flow pathways through mineral soils. Rocky sites tended to have much larger within-site variability in soil-water retention, relative to silty sites (Figure 5). The West Twin location, which is classified as rocky, has particularly large variability in soil-water retention (Figure 5). Tracer experiments by Koch et al. (2017) at the West Twin location suggested tracer transport into deeper preferential flow paths via discrete gravel regions. Prior

Table 6
Summary of Root Measurements in Mineral Soil From Soil Pits and Root Presence in Soil Cores

Site name	Abbreviation	Rocky/silty catchment-scale classification	Wildfire condition	Typical root diameter (mm)	Root penetration depth (cm)
West Fork of Dall burned	WFD B	Rocky	Burned	0.5 to 1	>100
West Fork of Dall unburned	WFD UB	Rocky	Unburned	<0.5	<55
West Twin unburned	WT UB	Rocky	Unburned	0.5 to 1	35
West Twin burned	WT B	Rocky	Burned	0.5 to 1	45
West Twin burned seep	WT BS	Rocky	Burned	0.5 to 1	55
Globe 1 burned	G1 B	Rocky	Burned	0.5 to 1	50
Globe 1 toeslope	G1 TS	Rocky	Unburned	0.5	<30
Globe 1 unburned	G1 UB	Rocky	Unburned	0.5	<30
Isom	Isom	Silty	Unburned	0.5 to 1	<50
Erickson burned upslope deciduous	E BUD	Silty	Burned	0.5 to 1	>115
Erickson burned	E B	Silty	Burned	0.5 to 1	100

Note. Sites are listed by the rocky and silty catchment classifications and then from north (first) to south (last).

work in boreal hillslopes has shown that preferential flow paths may develop because of soil texture spatial variability and root holes (Koch et al., 2014; Koch, Ewing, et al., 2013; Laine-Kaulio et al., 2015). Numerical modeling of unsaturated flow in nonpermafrost affected regions has shown that the spatial distribution of soil-water retention variability dominates the spatial evolution of preferential pathways, while K_{fs} variability has a smaller impact (Oliveira et al., 2006). The development of preferential flow pathways in response to soil variability can also be saturation dependent, with coarser soil textures providing preferential flow pathways under wetter conditions and finer soil textures serving as preferential pathways in dry conditions (Botros et al., 2012; Harter & Yeh, 1998; Roth, 1995). Based on the data collected here and comparison with prior field- and model-based studies, rocky sites may have greater

potential for subsurface preferential flow development in response to soil-water retention variability in mineral soil and therefore may have greater potential for preferential transport of water and solutes in boreal hillslopes. Fast pathways via preferential flow may have important implications for the mobilization of permafrost carbon given potentially relatively high permafrost dissolved organic carbon biodegradability (Abbott et al., 2014), although recent work has shown substantial variability in dissolved organic carbon biodegradability (Wickland et al., 2018). The rocky upland areas cover 30% of the areal extent of the boreal forest region of Alaska (Nossov et al., 2013) suggesting preferential flow as a potentially important mechanism in this region and highlighting the importance of understanding dominant mechanisms for water and solute movement in these regions.

4.3. Comparison to Prior Measurements, Implications, and Limitations

Comparison of the data from this work to prior work (Figures 9a-9c) shows that ranges of rocky and silty mineral soil values of ρ_b , K_{fs} , and soil-water retention broadly cover and considerably extend the parameter values from previous studies. Some studies noted here are outside of interior Alaska, such as the Hinzman et al. (1991) data from the North Slope, the Carey and Woo (2001) data from Canada, and the Yoshikawa and Hinzman (2003) data from the western coast of Alaska but were included for mineral soil comparisons because so little data are available in interior Alaska. Most notably, this work expands the ρ_b measurements range particularly toward lower values (Figure 9a), the coverage of the full spectrum of K_{fs} (Figure 9b), and the extension of soil-water retention into higher s values (Figure 9c). The sensitivity of

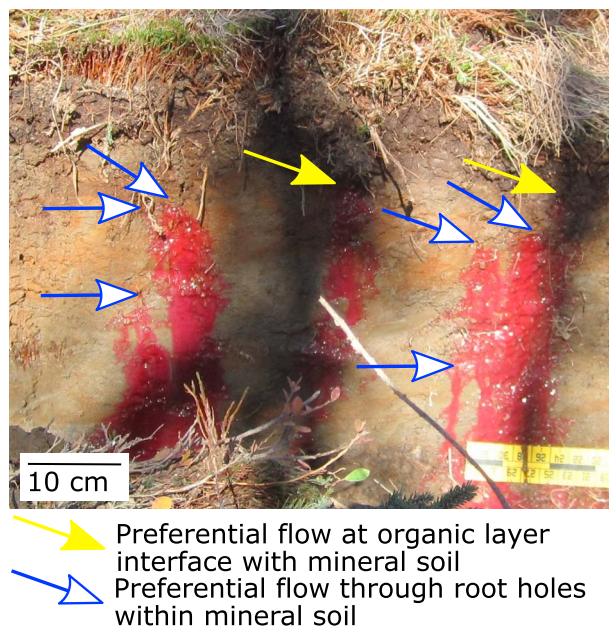
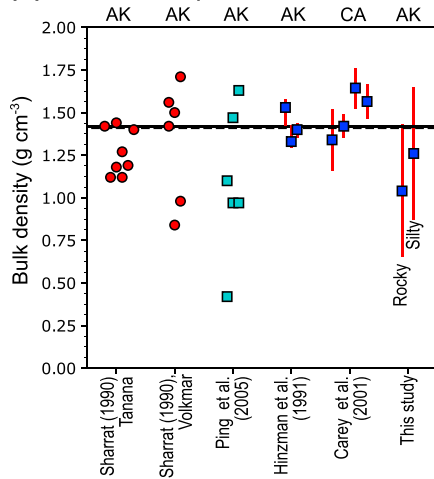


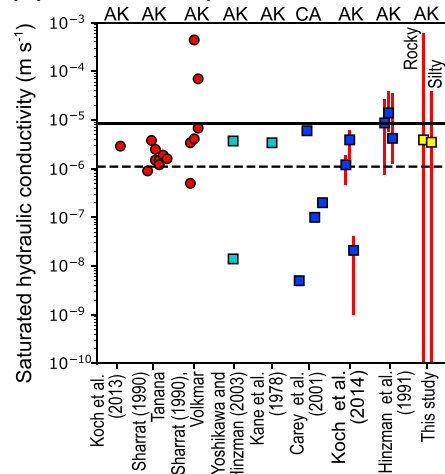
Figure 8. Photograph of preferential flow observed during Rhodamine dye tracing experiments at the West Fork of Dall burned site. Dye was applied via ponded water at the interface of the organic layer and mineral soil. Preferential flow was observed at the interface between the organic layer and mineral soil and also within root pathways within the mineral soil. The vertical flow down the trench face happens after preferential flow paths intersect the trench that was cut to observe subsurface dye movement.

--- Geometric mean (all data) ● Individual value
 — Arithmetic mean (all data) ■ Arithmetic mean
 ■ Arithmetic mean $\pm$ range or st. dev. ■ Geometric mean

(a) Bulk density



(b) Saturated hydraulic conductivity



(c) Soil-water retention

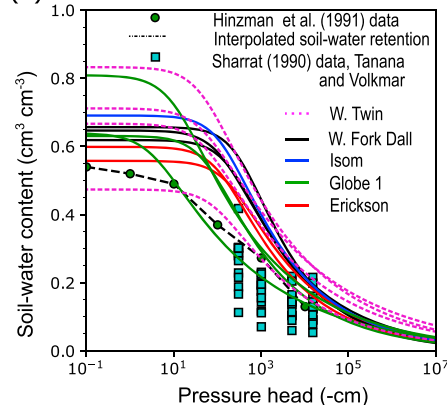


Figure 9. Comparison of the work presented here and selected prior studies focused on soil physical and hydraulic properties in Alaska (denoted as AK at the top of the figure) and similar regions in Canada (denoted as CA at the top of the figure): (a) bulk density, (b) saturated hydraulic conductivity, and (c) soil-water retention.

cryohydrogeologic models to mineral soil residual saturation (Harp et al., 2015) highlights the importance of the r values equal to zero at all sites (Table 4), which is well below typical minimum values (Harp et al., 2015). The range of K_{fs} covered in this work extends above 10^{-5} m/s, a value used as an approximate threshold above which groundwater flow should be considered for evaluating permafrost thaw due to the potentially important contribution of advection (McKenzie & Voss, 2013; Sjöberg et al., 2016). The geometric mean K_{fs} is greater than 10^{-5} m/s mostly at burned sites. Active layer dynamics following wildfire were recently shown to depend critically on organic and mineral soil K_{fs} , relative to other parameters, because of advective permafrost thaw by groundwater flow (Zipper et al., 2018). This work highlights the presence of measured mineral soil K_{fs} values at nearly 40% (5 out of 13) of the sites, with four out of the five the sites in the rocky classification, in the range of magnitudes shown to have the potential for substantial permafrost thaw via advective heat transport given groundwater flow (Lamontagne-Hallé et al., 2018; McKenzie & Voss, 2013; Sjöberg et al., 2016; Walvoord et al., 2019; Zipper et al., 2018). Detailed and extended site-based cryohydrogeologic modeling efforts are critical for elucidating the relative importance of advective and conductive thaw of permafrost across the full spectrum of boreal landscapes and ecosystem types. Data and soil property relationships presented here provide a foundation for such analysis.

While the soil hydraulic property measurements presented here cannot always be directly used in numerical models because of scale effects, they can provide bounds for effective parameter values and set realistic ranges for uncertainty estimation. The impact of large macropores is likely not accurately included in either the K_{fs} or soil-water retention measurements because of the soil core size. The thermal property and physical property measurements at the core scale may better reflect field scale parameters for use in modeling. While inverse parameter modeling using observed data may be a more accurate estimation method for soil hydraulic properties, the observed data needed to drive (e.g., climate and snow properties) and evaluate (soil temperature and saturation) are sparse in high latitudes (Chadburn et al., 2015) signifying the reliance on and added value of core-scale measurements. The data presented here are from a limited geographic area of 13 sampling sites at five locations; however, the range of K_{fs} , s , and bulk density covers and even exceeds the range of prior measurements in analogous locations, highlighting the importance of this data set.

5. Conclusions

Analysis of soil physical and hydraulic properties for sites in boreal forest regions of interior Alaska shows the following:

1. Silty and rocky classifications of upland soil landscapes for interior Alaska proved useful for classifying field areas and overall showed moderate to strong evidence of differences for mineral soil ρ_b , soil organic matter, sand fraction, silt fraction, and gravel fraction. Thermal properties of mineral soil showed substantial deviations in k compared to commonly used estimation methods for rocky sites where large gravel fractions were present. Soil organic matter variations were the clearest predictor of ρ_b and θ_s , while ρ_b was the clearest predictor of

- K_{fs} . The van Genuchten parameters α and N that describe soil-water retention relations depended strongly on clay and sand fractions but did not depend as strongly on ρ_b and soil organic matter.
- There was little evidence of a direct wildfire effect on soil physical or hydraulic properties. This finding agrees with prior work showing that wildfire in boreal forest areas of interior Alaska typically does not impart a substantial heat impulse into soil because of the thermal buffering of thick organic layers. Hydraulic conductivity values showed moderate evidence for greater variability at burned sites. The K_{fs} variability corresponded to observations of deep root penetration and root presence in soil cores and possible preferential flow during K_{fs} measurement. This link provides the basis for a hypothesis that deepening of active layers and deciduous vegetation leads to vertical root growth to depth and associated preferential flow that could facilitate permafrost thaw by advective groundwater flow-driven thaw mechanisms. Measured K_{fs} extends above 10^{-5} m/s, a value used as an approximate threshold for the onset of substantial permafrost thaw by advective heat transport via groundwater flow, at nearly 40% of sites. Cryohydrogeologic modeling will be useful in further exploring this hypothesis.
 - The characterization of soil physical and hydraulic properties provides a foundation for parameterizing cryohydrogeologic models to test hypotheses with numerical experiments that may illuminate climate and wildfire disturbance impacts on permafrost thaw and coupled hydrologic and biogeochemical fluxes in silty and rocky upland soil landscapes in interior Alaska.

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